MSc Technologies and Analytics in Precision Medicine

Module: Advanced Data Analytics

Assignment 5

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1 Introduction

This analysis investigates the relationship between blood glucose status and 30-day hospital readmission among patients with diabetes using observational hospital data. In observational datasets, patients are not randomly assigned to exposure groups, meaning that individuals with high blood glucose may differ systematically from those with normal levels. These differences in baseline characteristics may influence both glucose status and the risk of readmission.

Variables that affect both the exposure and the outcome are known as confounders, and they can distort the observed association between exposure and outcome. As a result, crude comparisons between patients with high and normal blood glucose may not accurately reflect the true relationship between glucose status and readmission.

To address this issue, propensity score methods can be used to adjust for measured confounding. Propensity scores estimate the probability of exposure conditional on observed covariates and can be used to create a weighted pseudo-population in which the distribution of covariates is more balanced between exposure groups.

The aim of this analysis is to evaluate the association between high blood glucose and 30-day hospital readmission and to assess the extent to which the observed relationship may be explained by measured confounding by comparing crude, regression-adjusted, and inverse probability of treatment weighted (IPTW) estimates.

2 Methods

2.1 Data Preparation

The dataset was restricted to patients with normal or high blood glucose, excluding other exposure categories and observations with missing blood glucose values. Missing values in categorical variables were recorded as a ”Missing” category to retain observations. Observations with missing continuous values were excluded from models. As a result of this, the dataset was reduced from 71,515 to 6,682 observations, meaning approximately 90.7% of the original data were removed.

2.2 Crude Association

A crude association between blood glucose status and 30-day readmission was estimated using logistic regression, with readmission as the outcome and blood glucose (high vs normal) as the exposure.

2.3 Assessing Confounding

Baseline characteristics were compared between exposure groups to identify potential confounders. Variables examined included age, sex, race, admission source, insurer type, duration of stay, number of medications, number of procedures, and number of diagnoses

2.4 Propensity Score Estimation

A propensity score model was fitted using logistic regression to estimate the probability of high blood glucose conditional on observed covariates. Propensity score distributions were summarised by exposure group and visualised using density plots to assess overlap.

2.5 IPTW Analysis

Inverse probability of treatment weights were calculated as $w = \frac{1}{PS}$ for exposed individuals and $w = \frac{1}{1-PS}$ for unexposed individuals. Weights above the 99th percentile were truncated to reduce the influence of extreme values

2.6 Modelling

Three models were compared: a crude logistic regression, a regression-adjusted model including all covariates, and a weighted logistic regression using IPTW. Odds ratios from these models were compared to evaluate the impact of confounding

3 Results and Interpretation

3.1 Crude Association

Table 1 presents the crude odds ratio for 30-day hospital readmission comparing patients with high blood glucose to those with normal blood glucose. The estimated odds ratio for high versus normal glucose is 1.00, with a 95% confidence interval ranging from 0.85 to 1.20.

Exposure	OR	2.5%	97.5%
High vs Normal Glucose	1.00	0.85	1.20

Table 1: Crude odds ratio for 30-day readmission comparing High versus Normal blood glucose

3.1.1 Interpretation

The crude odds ratio indicates no association between high blood glucose and 30-day readmission. However, this estimate may be unreliable due to confounding by indication. Patients with elevated glucose levels may differ systematically from those with normal levels in terms of

co-morbidity burden, length of hospital stays, and intense treatment, all of which independently influence readmission risk. Because exposure status is not randomly assigned in observational data, the crude comparison may not reflect the true causal effect of high blood glucose.

3.2 Assessing Confounding

Table 2 presents the baseline characteristics of patients with normal and high blood glucose levels. Categorical variables are reported as counts and percentages, and continuous variables are summarised as mean (standard deviation). The distribution of race, age, admission source, and insurer differed significantly between the two glucose groups (all $p < 0.001$). No statistically significant differences were observed for sex, duration of stay, number of diagnoses, number of procedures, number of medications, or number of previous visits.

Characteristic	Normal (N=3,791)	High (N=2,891)	p-value
Race			< 0.001
African American	810 (21%)	410 (14%)	
Asian	27 (0.7%)	28 (1.0%)	
Caucasian	2,703 (71%)	2,243 (78%)	
Hispanic	107 (2.8%)	77 (2.7%)	
Other	70 (1.8%)	61 (2.1%)	
Unknown	74 (2.0%)	72 (2.5%)	
Sex			0.060
Female	2,031 (54%)	1,481 (51%)	
Male	1,760 (46%)	1,410 (49%)	
Age			< 0.001
[0–10]	9 (0.2%)	2 (< 0.1%)	
[10–20]	35 (0.9%)	9 (0.3%)	
[20–30]	69 (1.8%)	23 (0.8%)	
[30–40]	162 (4.3%)	70 (2.4%)	
[40–50]	450 (12%)	263 (9.1%)	
[50–60]	661 (17%)	484 (17%)	
[60–70]	770 (20%)	671 (23%)	
[70–80]	897 (24%)	789 (27%)	
[80–90]	644 (17%)	506 (18%)	
[90–100]	94 (2.5%)	74 (2.6%)	
Admission Source			< 0.001
Emergency	2,556 (67%)	1,820 (63%)	
Other	335 (8.8%)	303 (10%)	
Referral	900 (24%)	768 (27%)	
Insurer			< 0.001
Medicaid	203 (5.4%)	75 (2.6%)	
Medicare	1,105 (29%)	922 (32%)	
Private	937 (25%)	652 (23%)	
Self-Pay	153 (4.0%)	147 (5.1%)	
Missing	1,393 (37%)	1,095 (38%)	
Duration (mean, SD)	4.9 (3.1)	4.8 (3.1)	0.443
Number of Previous Visits (mean, SD)	0.39 (1.08)	0.44 (1.30)	0.128
Number of Diagnoses (mean, SD)	7.52 (1.90)	7.54 (1.83)	0.620
Number of Procedures			0.293
0	1,894 (50%)	1,451 (50%)	
1	628 (17%)	453 (16%)	
2	436 (12%)	304 (11%)	
3	359 (9.5%)	300 (10%)	
4	151 (4.0%)	109 (3.8%)	
5	110 (2.9%)	86 (3.0%)	
6	213 (5.6%)	188 (6.5%)	
Number of Medications (mean, SD)	16 (9)	17 (9)	0.081

Table 2: Baseline characteristics of patients with Normal and High blood glucose levels. Values are n (%) unless otherwise stated.

3.2.1 Interpretation

Several baseline characteristics differed significantly between exposure groups and may confound the relationship between blood glucose and 30-day readmission. Age is strongly associated with glucose intolerance and abnormal glucose metabolism, with elderly individuals more likely to demonstrate elevated post-challenge glucose levels and abnormal glucose status (Kalyani and Egan 2013). Additionally, patients aged over 60 years have been identified as high-risk for hospital readmission due to more complex health needs (Sanseverino et al. 2025). Age therefore satisfies the criteria for confounding, as it is plausibly associated with both exposure and outcome.

Race may also confound the association. Evidence indicates that non-white patients with diabetes have a higher risk of 30-day readmission compared to white patients, and disparities in diabetes monitoring and healthcare quality have been reported across racial groups (North Carolina Healthcare Association 2022). As racial differences are associated with both glucose status and readmission risk, race may distort the crude exposure–outcome relationship.

Admission source may reflect severity of illness at presentation. Higher blood glucose levels on admission have been associated with increased short-term readmission and mortality (Evans and Dhatariya 2012). Patients admitted emergently or with acute metabolic disturbance may therefore be more likely to exhibit elevated glucose and subsequently experience readmission, making admission source a potential confounder.

Finally, insurer type may capture socioeconomic status and access to healthcare resources. Socioeconomic disparities and insurance coverage have been associated with differences in diabetes management and adverse outcomes, partly mediated through access to technology and care (Everett and Wisk 2021). As insurer type may influence both glucose control and risk of readmission, it may confound the observed association.

3.3 Propensity Score Estimation

Table 3 summarises the distribution of estimated propensity scores for patients with normal and high blood glucose levels. The mean propensity score was slightly higher among patients with high glucose (0.446) compared with those with normal glucose (0.423). *Figure 1* shows the density distribution of propensity scores for both groups. The distributions span similar ranges, with substantial overlap between groups.

Blood Glucose	n	Mean	SD	Min	P25	Median	P75	Max
Normal	3791	0.423	0.081	0.083	0.378	0.442	0.478	0.595
High	2891	0.446	0.065	0.133	0.417	0.457	0.489	0.604

Table 3: Summary statistics of the estimated propensity scores by blood glucose group.

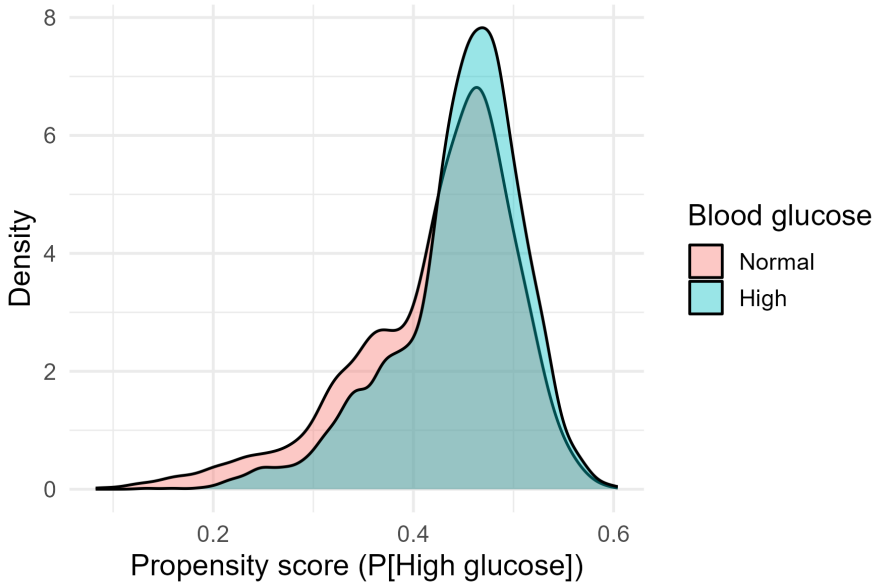

Figure 1: Density plot showing the distribution of estimated propensity scores for patients with Normal and High blood glucose levels.

3.3.1 Interpretation

The propensity score represents the probability of receiving the exposure given the observed covariates, in this case the probability of having high blood glucose conditional on demographic and clinical characteristics. As several of these variables may influence both exposure and outcome, they act as potential confounders.

Propensity scores address this by compressing multiple confounders into a single scalar probability representing the likelihood of exposure given the observed covariates. Individuals with similar characteristics therefore have similar propensity scores. By using these scores in the weighting procedure, the distribution of confounders can be made more balanced between exposure groups, mimicking the conditions of randomisation in observational data.

Adequate overlap in propensity score distributions between exposure groups is required for valid comparisons. Overlap indicates that patients with similar covariate profiles appear in both groups and therefore have comparable probabilities of receiving either exposure. In this situation, comparisons between groups are made among individuals with similar characteristics, reducing the influence of confounding when estimating the relationship between blood glucose and readmission.

3.4 IPTW Analysis

Table 4 shows the distribution of inverse probability treatment weights (IPTW) before and after truncation. IPTW weights ranged from 1.09 to 7.50 before truncation. Weights above the 99th percentile (3.58) were truncated to reduce the influence of extreme observations. This reduced

the maximum weight while leaving the overall distribution largely unchanged.

Weights	Min	1st Quartile	Median	Mean	3rd Quartile	Max
IPTW (before truncation)	1.09	1.76	1.95	2.00	2.16	7.50
IPTW (after truncation)	1.09	1.76	1.95	1.99	2.16	3.58

Table 4: Distribution of inverse probability of treatment weights before and after truncation at the 99th percentile.

Table 5 displays the odds-ratios of for the crude, regression-adjusted and IPTW models. The estimated association between high blood glucose and 30-day readmission was similar across modelling approaches, with crude (OR = 1.00), regression-adjusted (OR = 1.03), and IPTW-weighted estimates (OR = 1.02) all indicating little evidence of an association

Model	Odds Ratio (High vs Normal glucose)
Crude model	1.00
Regression-adjusted model	1.03
IPTW model	1.02

Table 5: Comparison of crude, regression-adjusted, and IPTW odds ratios for the association between high blood glucose and 30-day readmission.

4 Discussion

The crude model estimated the association between blood glucose status and 30-day readmission without adjusting for patient characteristics. In observational data such estimates may be biased due to confounding, as some variables can influence both exposure and outcome. The regression-adjusted model attempts to address this by including these potential confounders directly in the outcome model.

Propensity score weighting addresses confounding through a different method. The propensity score compresses multiple confounders into a single probability representing the likelihood of having high blood glucose given the observed characteristics. Individuals are then weighted according to the inverse probability of their exposure status. Patients who received an exposure that was unlikely given their characteristics receive larger weights, while those whose exposure was more expected receive smaller weights. This weighting creates a pseudo-population in which patients with similar covariate profiles are more equally represented across exposure groups. As a result, the exposure becomes approximately independent of the measured confounders, allowing the relationship between blood glucose and readmission to be estimated with reduced confounding.

In this analysis the crude, regression-adjusted and IPTW estimates were all similar, with odds ratios close to one. This suggests that the measured confounders had little influence on the crude

association and that both adjustment methods produced consistent estimates. Taken together, these results provide little evidence of an association between high blood glucose and 30-day readmission in this dataset.

However, these findings should be interpreted cautiously. Propensity score methods can only adjust for confounding from variables that were measured and included in the analysis, and unmeasured confounding may still be present. Additionally, covariate balance after weighting was not formally assessed using standardised mean differences (SMD), which are commonly used to evaluate whether IPTW has successfully balanced baseline characteristics between exposure groups. In addition to this, the substantial reduction in the sample size may have reduced statistical power, potentially limiting the ability to detect smaller associations.

The causal interpretation of these results could be strengthened by assessing balance diagnostics and by incorporating additional clinical variables that may influence both glucose status and readmission risk include indicators of diabetes management and disease burden, such as haemoglobin A1c levels, insulin treatment prior to admission, and the presence and severity of comorbidities measured using indices such as the Charlson Comorbidity Index (McDaniel and Chou 2022).

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